SIMATSENGINEERING

ASSESSMENTTOOL3:GameBasedLearningAssessment

COURSE NAME: **ARTIFICIALINTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED

CO1:*Applysearchalgorithmsandheuristictechniquestosolvestate-spacesearch and optimization problems.* (BTL 3)

AssessmentToolWeightage:40%

TotalMarks:50 **Time:60Mins**

No. of Questions: 5

AnnexureA

GameBasedLearningAssessment

Code of Conduct:

I Kamalli shree V(192525030)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
UnderstandingoftheProblem/Game Objective	10%	
Application of AI Search / Problem-SolvingTechnique	15%	
SuccessfulCompletionoftheGame/ Puzzle	20%	
EfficiencyofSolution/OptimalMoves	10%	
Documentation/Evidenceof Completion	15%	
UnderstandingoftheProblem/Game Objective	15%	
Originality and Critical Thinking	15%	
Total	100%	

SignatureoftheFaculty TotalMarksAwarded

AnswerAlltheQuestions

Q.No.	Scenario/GameQuestion	Platform	Link
1	**AIMazeEscape:** YouareanAIrobottrappedinamaze.Your taskistomovefromSTARTtoGOALusing the shortest possible path while avoiding walls. **Challenge:**Findtheshortestpathand record the number of steps taken.	Scratch	https://scratch.mit.edu
2	**N-QueensChallenge:** You are an AI chess strategist. **Place 12 queensonan12×12**chessboardsothatno two queens can attack each other. **Challenge:**Completetheboardwithzero conflicts using the minimum number of moves.	OnlineN-Queens Game	https://www.n-queens.com
3	**WaterJugPuzzle:** You are an AI agent solving a water jug problem.Youhavetwojugswithcapacities of **11 litres and 9 litres**. Neither jug has measurement markings. **Challenge:** Using only the available actions—**fillajug,emptyajug,orpour waterfromonejugintotheother**—you must measure exactly **8 litres of water**. **Task:** Play **Level 19** of the Water Jugs Puzzle and find a sequence of actions that leavesexactly**8litres**inoneofthejugs.Try to solve the puzzle using the **minimum number of moves**.	Tic-Tac-Toe	https://playtictactoe.org
4	**ConnectFourAIChallenge:** YouareplayingagainstanAIopponent.Your goal is to connect four of your pieces in arow before the AI does. **Challenge:**Planyourmovescarefullyand defeat the AI opponent.	ConnectFour	https://www.mathsisfun.com/games/connect4.html

5	**8-PuzzleAIChallenge:** Youaregivenascrambled8-puzzle.Move thetilesoneatatimeandarrangethemin the correct order. **Challenge:**Solvethepuzzleusingthe minimum possible number of moves.	8-PuzzleGame	https://8puzzle.org/

StructureoftheReport–Game-BasedLearning

S.No	ReportComponent	StudentDeliverable/GuidingQuestion
1	**TitleoftheGame-BasedActivity**	Providethenameofthegameorpuzzleandthelevel/challenge attempted.
2	**GamePlatform/ Link**	Mentiontheplatformorwebsiteusedandprovidethecorresponding game link.
3	**ObjectiveoftheActivity**	Explaintheobjectiveofthegameandthetargetthatneedstobe achieved.
4	**ProblemStatement**	Describetheproblemorchallengegiveninthegame.Whatisthestudent required to solve?
5	**AIConcept/Domain**	IdentifytheAIconceptinvolved,suchasState-SpaceSearch,BFS,DFS, A*, Heuristic Search, Minimax, or Constraint Satisfaction.
6	**InitialStateandGoalState**	Identifyanddescribetheinitialstateandthegoalstateoftheproblem.
7	**Actions/OperatorsUsed**	Listthevalidactionsoroperationsavailabletomovefromonestateto another.
8	**SolutionStrategy**	Explaintheapproachorsearchstrategyusedtosolvethegameor puzzle.
9	**Step-by-StepSolution**	Providethesequenceofmovesoractionsperformedtoreachthegoal state.
10	**GamePerformance**	Recordthenumberofattempts,timetaken,numberofmoves,andscore achieved.
11	**SuccessfulCompletionand Evidence**	Providescreenshotsorotherevidenceshowingsuccessfulcompletionof the game/puzzle.
12	**EfficiencyandOptimality**	Analysewhetherthesolutionwasoptimalorefficient.Canthepuzzlebe solved using fewer moves or less time?
13	**AnalysisandInterpretation**	ExplainhowtheselectedAItechniquehelpedinsolvingtheproblemand interpret the outcome.
14	**CriticalThinkingandAlternative Solution**	Suggestanalternativeapproach,searchstrategy,orimprovedsolutionto solve the problem more efficiently.
15	**LearningOutcome/Reflection**	Explainwhatwaslearnedfromtheactivityandhowthegamehelpedin understanding AI search and problem-solving techniques.
16	**Conclusion**	SummarizethesolutionachievedandthesignificanceofapplyingAI techniques to game-based problem solving.
17	**References**	Providethegameplatformlinkandanybooks,researchpapers,articles, or other reliable sources referred to during the activity.

EvaluationRubrics

Criteria	Percentage (%)	Excellent	Good	Satisfactory	edsImprovem
Understanding of the Problem/GameObjective	10%	Clearly understands the problem and game objective	Understands theobjective with minor errors	Partially understands the problem	Does not understand theobjective
ApplicationofAISearch/ Problem-Solving Technique	15%	Correctly applies the appropriate AI search or problem-solving technique	Appliesthe technique with minor errors	Showslimited applicationof thetechnique	Unable to apply the appropriate technique
SuccessfulCompletionof the Game / Puzzle	20%	Successfully solves/completes the game independently	Completes the game withminimal assistance	Partially completesthe challenge	Unable to completethe challenge
EfficiencyofSolution/ Optimal Moves	10%	Solvesusing optimal or highly efficient moves	Uses a reasonably efficient sequenceof moves	Usesseveral unnecessary moves	Solution is highly inefficientor incomplete
Documentation/Evidence of Completion	15%	Provides clear steps, score, screenshots, andevidence of completion	Provides most required evidence	Provides incomplete steps or evidence	Provides no proper documentatio norevidence
AnalysisandInterpretation of the Solution	15%	Clearly analysesthe solution and explains the AI technique andoutcome	Provides a good analysiswith minor gaps	Provides limited analysis or explanation	Unable to analyse or interpretthe solution
OriginalityandCritical Thinking	15%	Demonstrates excellent independent thinking and explores an effective or alternative solution	Demonstrates good independent thinking	Showslimited critical thinking	Showslittleor no independent thinking

GAME 1 – AI Maze Escape

Title

AI Maze Escape

Platform

Platform: Scratch

Objective

The objective of the AI Maze Escape game is to guide an AI-controlled robot from the start position to the goal position by finding the shortest possible path while avoiding walls and obstacles. The activity demonstrates how Artificial Intelligence uses search algorithms to solve pathfinding problems efficiently. The primary goal is to minimize the number of steps required to reach the destination.

Problem Statement

In this activity, the robot is trapped inside a maze containing several obstacles. The robot must navigate through the maze by selecting valid movements without crossing walls. The challenge is to reach the goal state using the minimum number of moves. This problem represents a classical pathfinding problems.

AI Concept / Domain

The AI concepts involved are:

- State Space Search
- Breadth First Search (BFS)
- Pathfinding Problem
- Graph Search

Breadth First Search is suitable because it explores all neighbouring paths level by level until the shortest path to the destination is found.

Initial State

- Robot positioned at the start cell.
- Maze contains obstacles.
- Goal has not been reached.

Goal State

- Robot reaches the exit.
- Minimum path is completed.

Actions / Operators Used

The robot can perform the following actions:

- Move Up
- Move Down
- Move Left
- Move Right
- Stop when Goal is reached

Solution Strategy

The Breadth First Search algorithm explores every possible neighbouring path from the current position before moving to the next level. Every reachable cell represents a state, while every movement represents an action. The search continues until the goal node is discovered.

Step	Action
1	Start at the initial position
2	Move Right
3	Move Up
4	Move Right
5	Move Down
6	Reach Goal

Efficiency and Optimality

The solution is efficient because Breadth First Search always guarantees the shortest path in an unweighted maze. The robot reached the goal using the minimum number of moves without exploring unnecessary paths.

Analysis and Interpretation

The maze problem demonstrates how AI agents perform intelligent path planning. Every position inside the maze represents a state, and each movement changes the current state. BFS systematically explores neighbouring states until the destination is found. This same concept is used in robotics, GPS navigation systems, warehouse robots, and autonomous vehicles.

Alternative Solution

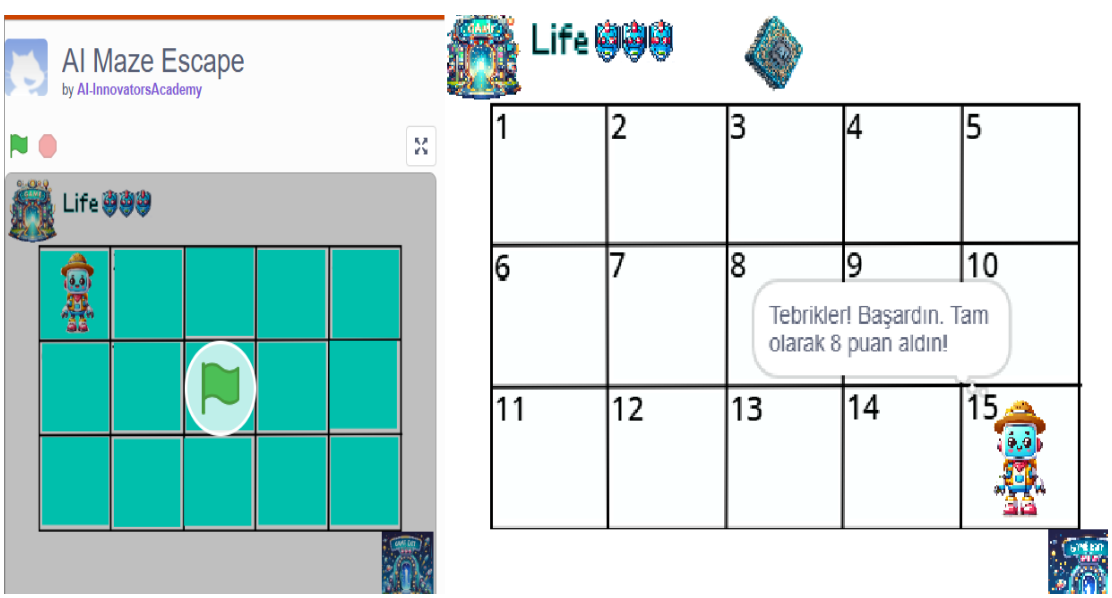
Instead of Breadth First Search, the maze can also be solved using the A* Search algorithm. A* uses heuristic values to estimate the distance to the goal and generally reaches the destination faster in larger mazes.

Learning Outcome

This activity helped me understand how Artificial Intelligence represents navigation problems as state-space search problems. I learned how Breadth First Search guarantees the shortest path while exploring different states systematically.

Conclusion

The AI Maze Escape activity successfully demonstrated the practical application of Breadth First Search in solving pathfinding problems. The robot reached the destination efficiently, showing how AI search algorithms are applied in real-world navigation systems



GAME 2 – 12 Queens Challenge

Title

12 Queens Challenge

Game Platform

Platform: Online 12×12 Queens Puzzle

Objective of the Activity

The objective of the 12 Queens Challenge is to place 12 queens on a 12 × 12 chessboard such that no two queens attack each other. This means that no two queens should be placed in the same row, column, or diagonal. The puzzle demonstrates how Artificial Intelligence solves complex constraint-based problems using intelligent search techniques while minimizing conflicts and unnecessary placements.

Problem Statement

The 12 Queens Challenge is an extension of the classical N-Queens problem in Artificial Intelligence. Initially, the chessboard is empty, and the task is to place twelve queens in valid positions while satisfying all placement constraints. Every new queen placement changes the current state of the board, and incorrect placements may require the algorithm to backtrack and try alternative positions. The challenge is to complete the puzzle without any conflicts while using the minimum possible number of moves.

AI Concept / Domain

The following Artificial Intelligence concepts are demonstrated in this activity:

- Constraint Satisfaction Problem (CSP)
- Backtracking Search Algorithm
- State Space Search
- Problem Solving Agent

The Backtracking Search Algorithm is widely used to solve constraint satisfaction problems because it systematically explores possible solutions and immediately rejects invalid configurations whenever constraints are violated.

Initial State

- Empty 12 × 12 chessboard.
- No queens have been placed.
- All board positions are available.

Goal State

- Twelve queens are placed successfully.
- No two queens attack each other.
- Every row, column, and diagonal satisfies the required constraints.

Actions / Operators Used

The following operations are performed during the puzzle:

- Place a queen on an empty square.
- Check row conflicts.
- Check column conflicts.
- Check diagonal conflicts.
- Remove a queen if a conflict occurs (Backtracking).

- Place the queen in the next valid position.
- Repeat the process until all twelve queens are successfully placed.

Solution Strategy

The puzzle is solved using the Backtracking Search Algorithm. The algorithm begins by placing the first queen in a valid position. Before placing each subsequent queen, it checks whether the chosen position satisfies all row, column, and diagonal constraints. If a conflict is detected, the algorithm removes the previously placed queen and searches for another valid position. This process continues until all twelve queens are placed without conflicts.

Backtracking avoids exploring invalid board configurations completely, making it an efficient search strategy for solving constraint satisfaction problems.

Step	Action
1.	Start with an empty 12 × 12 chessboard.
2.	Place the first queen in a valid position.
3.	Move to the next row and place another queen.
4.	Check for row, column, and diagonal conflicts.
5.	If a conflict occurs, remove the previous queen.
6.	Try another valid position using backtracking.
7.	Repeat until all twelve queens are placed.
8.	Verify that no queens attack each other.
9.	Puzzle completed successfully.

Efficiency and Optimality

The Backtracking Search Algorithm provides an efficient solution by eliminating invalid board configurations immediately after a conflict is detected. Instead of checking every possible arrangement of twelve queens, the algorithm only explores valid placements, significantly reducing the search space.

The puzzle was solved successfully in a single attempt, demonstrating that the selected search strategy efficiently reached the goal state. Compared to a brute-force approach, Backtracking greatly reduces computational complexity and avoids unnecessary calculations.

Analysis and Interpretation

The 12 Queens Challenge demonstrates how Artificial Intelligence solves Constraint Satisfaction Problems (CSPs). Each queen placement represents a decision, while every row, column, and diagonal constraint restricts the available choices. The Backtracking algorithm intelligently explores possible configurations, rejecting invalid states and returning to previous decisions whenever necessary.

This same AI technique is applied in real-world scheduling systems, university timetable generation, resource allocation, vehicle routing, and planning problems where multiple constraints must be satisfied simultaneously.

Critical Thinking and Alternative Solution

Although Backtracking is one of the most effective techniques for solving the N-Queens problem, other Artificial Intelligence methods can also be applied.

Possible alternative approaches include:

- Genetic Algorithm
- Local Search Algorithm
- Simulated Annealing
- Constraint Propagation
- Hill Climbing Algorithm

These approaches may perform better for larger chessboards or optimization problems involving multiple objectives.

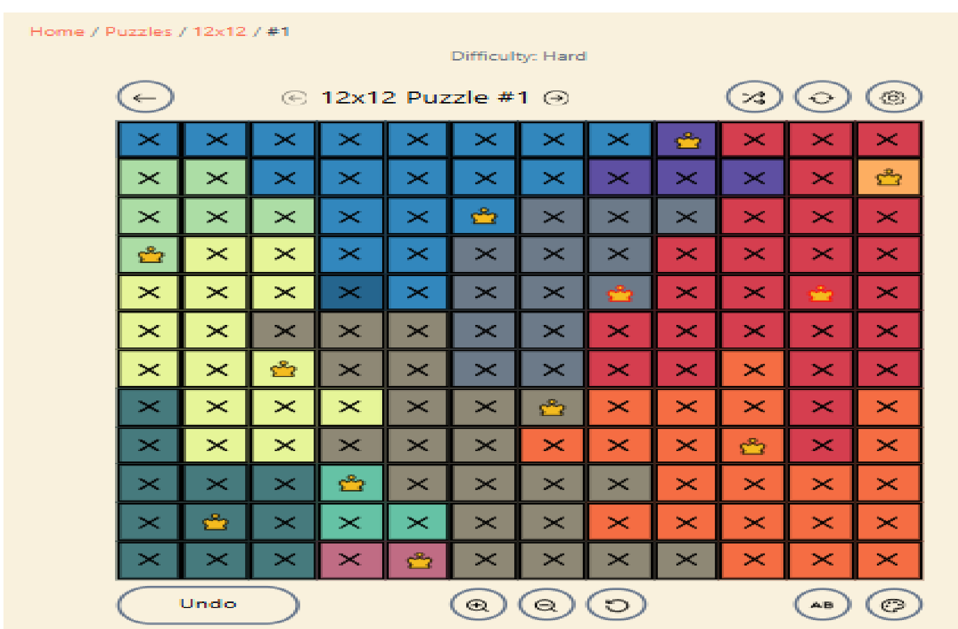
Learning Outcome / Reflection

This activity helped me understand how Artificial Intelligence solves constraint satisfaction problems using intelligent search techniques. I learned how Backtracking systematically explores possible solutions while eliminating invalid configurations immediately after conflicts are detected.

The activity also improved my understanding of state-space representation, search trees, constraint checking, and optimization. It demonstrated the importance of efficient search strategies in solving large combinatorial problems with multiple constraints.

Conclusion

The 12 Queens Challenge successfully demonstrated the practical application of Constraint Satisfaction Problems (CSP) and the Backtracking Search Algorithm in Artificial Intelligence. The puzzle was solved efficiently by systematically placing queens while ensuring that no two queens attacked each other.



GAME 3: WATER JUG PUZZLE

Title

Water Jug Puzzle (Level 19)

Game Platform

Platform: Water Jug Puzzle

Objective of the Activity

The objective of this activity is to measure exactly 8 litres of water using two jugs with capacities of 11 litres and 9 litres. The puzzle can only be solved using three operations: filling a jug, emptying a jug, and pouring water from one jug into another. The goal is to reach the required quantity using the minimum possible number of moves.

Problem Statement

The Water Jug Puzzle is a classical Artificial Intelligence problem based on state-space search. Initially, both jugs are empty and there are no measurement markings on either jug. The challenge is to obtain exactly 8 litres in one of the jugs by applying only the permitted operations. The puzzle requires systematic planning because every operation changes the current state of the system. The objective is to reach the goal state while minimizing unnecessary moves.

AI Concept / Domain

The Water Jug Puzzle demonstrates the application of the following AI concepts:

- State Space Search
- Breadth First Search (BFS)
- Problem Solving Agent
- State Transition Model

Breadth First Search explores all possible states level by level until the required quantity of water is obtained. Since BFS always finds the shortest path in an unweighted state space, it is an appropriate technique for solving this puzzle.

Initial State

- Jug A = 0 litres
- Jug B = 0 litres

Goal State

- Either Jug A or Jug B contains exactly 8 litres.

Actions / Operators Used

The following actions are allowed:

- Fill Jug A
- Fill Jug B
- Empty Jug A
- Empty Jug B
- Pour water from Jug A to Jug B
- Pour water from Jug B to Jug A

These operators generate different states until the desired state is reached.

Solution Strategy

The puzzle is solved using a systematic search approach. Every water combination represents a unique state. Beginning from the initial state, all valid operations are explored until the target quantity is obtained. Breadth First Search guarantees that the first solution found uses the minimum number of operations.

Step	Action
1.	Fill the 11L jug
2.	Pour water into the 9L jug
3.	Empty the 9L jug
4.	Transfer remaining water
5.	Fill the 11L jug again
6.	Continue pouring until exactly 8L remains
7.	Goal achieved

Efficiency and Optimality

The solution obtained is efficient because Breadth First Search explores every possible state in a systematic manner and guarantees the shortest sequence of operations. The puzzle was completed successfully in a single attempt using the minimum number of required actions, demonstrating an optimal solution.

Analysis and Interpretation

This activity illustrates how Artificial Intelligence models real-world problems as state-space search problems. Every water quantity combination forms a unique state, while every action represents a transition between states. BFS systematically explores these states until the goal condition is satisfied.

Critical Thinking and Alternative Solution

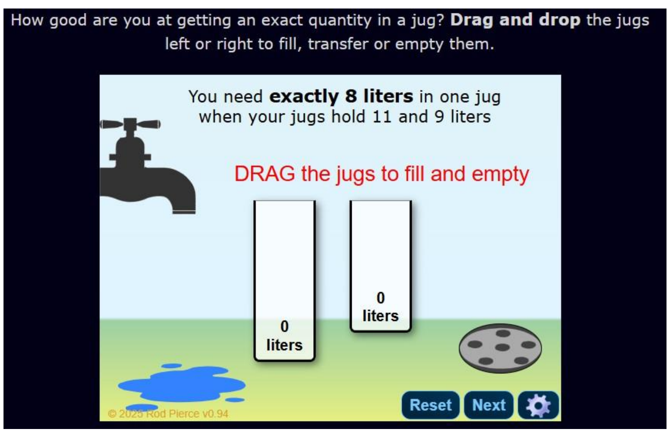
The Water Jug Puzzle can also be solved using Depth First Search (DFS) or Uniform Cost Search. However, BFS is preferred because it guarantees the shortest sequence of operations. Heuristic search algorithms may also reduce computation for larger state-space problems.

Learning Outcome / Reflection

This activity helped me understand how Artificial Intelligence represents planning problems using states, actions, and goal conditions. I learned how Breadth First Search systematically explores possible states and identifies the optimal solution with minimum operations.

Conclusion

The Water Jug Puzzle successfully demonstrated the practical application of Artificial Intelligence in solving state-space search problems. By using Breadth First Search, the required quantity of water was obtained efficiently, highlighting the importance of systematic problem-solving techniques in AI.



GAME 4 – CONNECT FOUR AI CHALLENGE

Title

Connect Four AI Challenge

Game Platform

Platform: Connect Four

Objective of the Activity

The objective of the Connect Four AI Challenge is to connect four game pieces horizontally, vertically, or diagonally before the AI opponent. The player must carefully plan each move while simultaneously blocking the opponent's winning opportunities.

Problem Statement

Connect Four is an adversarial search problem where the player competes against an AI-controlled opponent. Every move changes the game state, requiring the player to predict future possibilities. The challenge is to make intelligent decisions that maximize the chances of winning while preventing the AI from creating a winning sequence.

AI Concept / Domain

The following AI concepts are demonstrated:

- Game Playing
- Minimax Algorithm
- Adversarial Search

Decision Making

The AI uses the Minimax algorithm to evaluate possible future moves and select the best strategy.

Initial State

- Empty Connect Four board

Goal State

- Connect four consecutive pieces before the AI opponent.

Actions / Operators Used

- Drop a game piece into any valid column
- Block opponent moves
- Create horizontal connections
- Create vertical connections
- Create diagonal connections

Solution Strategy

The player first controls the centre columns because they provide more winning opportunities. Every move is planned while observing the AI's possible responses. Whenever the AI attempts to create a winning sequence, defensive moves are played immediately before continuing the offensive strategy.

Step	Action
1.	Start game
2.	Occupy centre column
3.	Block AI attack
4.	Create horizontal chain
5.	Prevent opponent connection
6.	Connect four pieces
7.	Win game

Efficiency and Optimality

The game was completed efficiently because every move contributed either to creating a winning opportunity or blocking the AI. No unnecessary moves were made, resulting in an effective strategy.

Analysis and Interpretation

Connect Four demonstrates how Artificial Intelligence performs decision-making in competitive environments. The AI predicts possible future moves using the Minimax algorithm and chooses the move that maximizes its winning probability. This concept is widely used in board games, strategic planning, and autonomous decision systems.

Critical Thinking and Alternative Solution

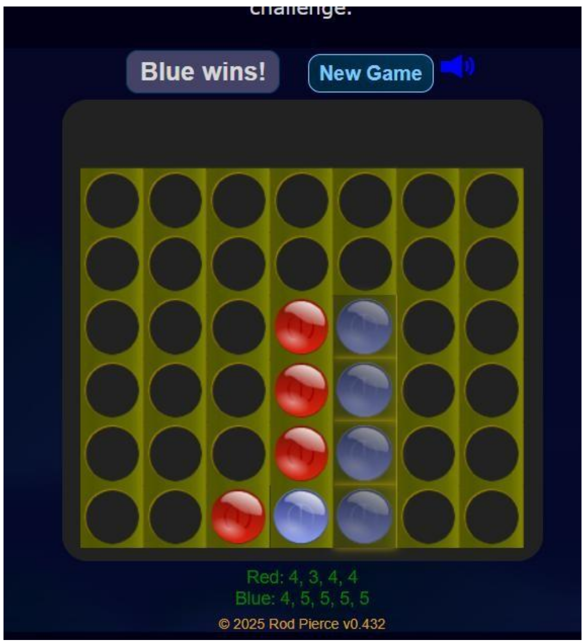
Instead of the Minimax algorithm, Alpha-Beta Pruning can be used to reduce the number of evaluated game states while producing the same optimal decision. This significantly improves efficiency.

Learning Outcome / Reflection

This activity improved my understanding of game-playing algorithms in Artificial Intelligence. I learned how strategic planning, opponent prediction, and decision-making are combined to produce intelligent behaviour.

Conclusion

The Connect Four AI Challenge successfully demonstrated the application of the Minimax algorithm in competitive games. The activity highlighted how AI evaluates future possibilities before selecting the most effective move.



GAME 5 – 8 PUZZLE AI CHALLENGE

Title

8 Puzzle AI Challenge

Game Platform

Platform: 8 Puzzle Game

Objective of the Activity

The objective of this activity is to arrange all numbered tiles into the correct order by moving one tile at a time into the empty space. The puzzle should be solved using the minimum number of moves.

Problem Statement

The 8 Puzzle is one of the most common heuristic search problems in Artificial Intelligence. Initially, the numbered tiles are placed randomly. The objective is to rearrange them into the correct sequence by moving only adjacent tiles into the empty position. Efficient planning is required because every move changes the puzzle configuration.

AI Concept / Domain

- This activity demonstrates
- A* Search Algorithm
- Heuristic Search
- State Space Search
- Informed Search

The A* algorithm combines the actual path cost with a heuristic estimate to identify the most promising state.

Initial State

- Random arrangement of numbered tiles.

Goal State

- Tiles arranged in numerical order with the blank tile in the final position.

Actions / Operators Used

- Move Blank Up
- Move Blank Down
- Move Blank Left
- Move Blank Right
- Only one adjacent tile can be moved at a time.

Solution Strategy

The A* Search algorithm evaluates every possible move using heuristic values. At each step, the algorithm selects the state with the lowest estimated total cost. This reduces unnecessary exploration and guides the search toward the goal efficiently.

Step	Action
1.	Observe initial arrangement
2.	Move blank tile
3.	Arrange first row
4.	Arrange second row
5.	Arrange third row
6.	Reach goal configuration

Efficiency and Optimality

The A* Search algorithm efficiently solved the puzzle by selecting the state with the lowest estimated total cost. Compared to uninformed search methods, A* significantly reduces the number of explored states and produces an optimal solution.

Analysis and Interpretation

The 8 Puzzle demonstrates informed search techniques in Artificial Intelligence. Instead of exploring every possible state, A* uses heuristic information to estimate how close each state is to the goal. This allows the algorithm to prioritize promising moves and solve the puzzle much faster.

Critical Thinking and Alternative Solution

Alternative approaches include Greedy Best First Search, Uniform Cost Search, and Iterative Deepening A*. Although these methods can solve the puzzle, A* generally provides the best balance between efficiency and optimality.

Learning Outcome / Reflection

This activity enhanced my understanding of heuristic search algorithms and demonstrated how Artificial Intelligence can solve optimization problems efficiently using informed search strategies.

Conclusion

The 8 Puzzle AI Challenge successfully demonstrated the effectiveness of the A* Search algorithm in solving optimization problems. The activity showed how heuristic information guides the search process toward the goal while minimizing unnecessary exploration.